HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Authored and maintained **MES recipes** and worksheets for a manufacturing site using **POMSnet**, directly translating process requirements into functional design.
- Designed, authored, and updated **MES recipes**, worksheets, and equipment specifications, including phase transition logic using **C#** and **VB Script** on the **.NET framework**.
- Provided technical training to site personnel on their respective **MES roles** and managed system access permissions, ensuring effective user adoption.
- Developed and executed **test and validation protocols**, risk assessments, and comprehensive system documentation in line with **cGMP** requirements.
- Integrated **POMSnet** with other site systems, including database development in **Microsoft SQL Server** and **Oracle** to support data flow and reporting.
- Collaborated with customer engineering teams to gather and translate project and process improvement requirements into detailed design documentation for **automation projects**.
- Containerized MES-related services using **Docker** and managed deployments on **AWS EC2** instances, ensuring reliable application availability.

SKILLS

MES Systems – POMSnet, Recipe Authoring, Worksheet Design, Phase Transition Logic, cGMP

Programming Languages – C#, VB Script, SQL, Python, Java

Development Frameworks – .NET Framework, Spring Boot

Databases – Microsoft SQL Server, Oracle, PostgreSQL, MySQL

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

DevOps & Tools – Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

PROJECTS

POMSnet Recipe Management System Enhancement

- Developed a modular system for authoring and managing complex **MES recipes** within **POMSnet**, improving the efficiency of recipe creation by an estimated **15%**.
- Implemented data validation routines using **C#** to ensure recipe integrity before deployment, reducing execution errors by approximately **20%**.
- Authored detailed technical documentation and user guides for the new recipe management system, facilitating knowledge transfer to the operations team.

Manufacturing Data Integration and Reporting

- Engineered ETL pipelines using **Python** and **SQL** to extract and transform manufacturing data from MES and ERP systems into a centralized data warehouse.
- Designed and built interactive **Power BI** dashboards to visualize key production metrics, enabling stakeholders to monitor performance and identify areas for improvement.
- Automated the generation of daily operational reports using **Python scripting**, reducing manual reporting time by over **5 hours per week**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)